Deadlock concept :→

(Server)
(API) Application Programming Interface

Bank Account A — https req — (lock) — https req — Bank Account B

User A tries to send some amount to user B. {Simultaneously} User B tries to send some amount to user A

{ Both of them send some request to each other & lock the system at the same time }

* When two or more threads try to access the same resource (simultaneously) the system freezes (Deadlock)

Arrays :→

An array is a fixed-size data structure that stores elements of the same data type in contiguous memory locations.

(elements)

2	8	1	6	4
0x1000	0x1004	0x1008	0x1012	0x1016
4B	4B	4B	4B	4B

Contiguous memory locations of equal size depending on the datatype. (int)

position → 1 2 3 4 5

int[] arr = { 2, 8, 1, 6, 4 };

index → 0 1 2 3 4

{ index = position − 1 ; position = index + 1 ; }

size of integer = 4 Bytes

0x1000	0x1001	0x1002	0x1003	0x1004
1B	1B	1B	1B	1B

char

* [Index based access]

4th element arr[3]

1st element arr[0]

nth element arr[n−1]

Key points :→

* Fixed size
* Index-based (starts from 0)
* Homogeneous data (same type)
* Stored in the heap memory (for objects)
* Faster access using index :

element n → arr[n−1]

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9
3	8	7	6

r, c

[0, 0] 1 (r c)

[0, 2] 3 (r c)

[3, 3] 6 (r c)

[r][c] = cell value

Transpose